MIL-C-50785\1B (AR)
15 July 1981
SUPERSEDING
MIL-C-50785/lA (MU)
1 September 1975

MILITARY SPECIFICATION

* CIRCUIT CARD ASSEMBLY: 9337291

(VOLTAGE COMPARATOR)

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. **SCOPE .**

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of Specification MIL-C-50785 comprise the complete requirements for the **Voltage Comparator Circuit Card Assembly.**

2. **APPLICABLE DOCUMENTS.**

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-50785	Circuit Card Assemblies, General Specification for

DRAWINGS **(See 6.2)**

US ARMY ARMAMENt RESEARCH AND DEVELOPmeNT **COMMAND (ARRADCOM)**

12000239	Sight Current Generator
9337291	Cirucit Card Assembly, Voltage Comparator

PACKAGING DATA SHEET

* 9337291	Packaging of Circuit Card Assembly, VOltage Comparator

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC: 1285

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:37Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTSs

3.1 Item definition. This assembly is a single ended printed circuit board consisting of three identical voltage comparators. The input to each comparator contains a resistor capacitor high frequency filter. The resistors also provide input current limiting. The assembly is used to compare input voltages when assembled as a part ot the Sight Current Generator Drawing Number 12000239.

3.1.1 Major component list. The components of this assembly shall be as specified on the Engineering Parts List (EPL) 9337291.

3.1.2 Government furnished Property list. Not applicable.

3.2 Characteristics.

3.2.1 Performance. The assembly shall be capable of comparing the relative voltage levels between two inputs and delivering an output voltage of either +15 Vdc or zero vdc, depending on which of tne input levels is dominant.

3.2.1.1 Input and output levels.

3.2.1.1.1 Input levels. The input power supplied to the **assembly shall be as specified in Table I.**

TABLE I. Input power

Pin	Voltage
4 and 25	+15 Vdc ± 2.0 percent
14	-15 Vdc ± 2.0 percent
1 and 28	Ground

3.2.1.1.2 Output levels. The requirements specified in 3.2.1.1.2.1 through 3.2.1.1.2.3 shall be met when checked with input voltages between and including -13 Vdc to +13 VdC. The source impedance shall be 6800 ohms ± 5 **percent.**

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:37Z
Check the source to verify that this is the current version before use.

3.2.1.1.2.1 Pin 2 output. The output shall be zero + 0.5 Vdc between pin two and pin one with an input voltage 35 millivolts more positive on pin 6 than on pin 5, and 15 +.1.0 Vdc between pin two and pin one with an input voltage 35 millivolts more negative on pin 6 than on pin 5.

3.2.1.1.2.2 Pin 26 output. The output shall be zero ± 0.5 Vdc between pin 26 and pin 28 with an input voltage 35 millivolts more positive-on pin 16 than on pin 15, and 15 ± 1.0 **VdC** between pin 26 and pin one with an input voltage 35 millivolts more negative on pin 16 than on pin 15.

3.2.1.1.2.3 Pin 27 output. The output shall be zero ± 0.5 Vdc between pin 27 and pin 28 with an input voltage 35 millivolt more positive on pin 23 than on pin 19, and 15 ± 1.0 Vdc between pin 27 and pin 28 with an input voltage 35 millivolts more negative on pin 23 than on pin 19.

3.2.1.2 Environmental. The assembly shall meet the environmental requirements of specification MIL-C-50785.

* 3.2.2 Physical characteristic. The physical characteristics snail be as specified in Drawing 9337291 and the engineering parts list.

3.3 Design and construction.

* 3.3.1 **Production drawings. This** assembly shall be fabricated in accordance with the requirements of specification MIL-C-50785, Drawing 9337291 and associated lists.

3.3.2 Standards of manufacture. The standards of manufacture shall be tnose specified in specification MIL-C-50785.

3.3.3 Workmanship. The workmanship shall be as specified in specification MIL-C-50785.

3.4 First article inspection. This specification makes provisions for First Article Inspection. Requirements for the submission of First Article Samples by the contractor shall be as specified in the contract.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:37Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS.

4.1 General.

4.1.1 Responsibility for inspection. The responsibility for inspection shall be as specified in Specification MIL-C-50785.

4.1.2 Special tests and examination. The assembly shall undergo first article and control testing in accordance with 4.1.2 of Specification MIL-C-50785.

4.2 Quality conformance inspections. The assembly shall be inspected and tested in accordance with 4.2 of Specification MIL-C-50785.

4.2.1. Test equipment. (See Figure 1)

4.2.1.1 Power supply **(Psl)**. The power supply shall be 15 ± 0.1 Vdc, 100 milliamperes (ma) or greater, ripple less than 10 millivolts (rev) rms or 100 Mv peak to **peak.**

4.2.L2 Power supply **(Ps2& The power supply shall be -15 ± 0.1 Vdc, 100 ma or greater, ripple less than 10** mv rms or 100 mv peak to peak.

4.2.1.3 Power supplies (PS3) and (PS4). These power supplies shall be zero ± 13 Vdc, with accuracy of adjustment and regulation to ± .01 percent, 100 ma or greater.

4.2.1.4 Resistors. Three resistors are required with a rating of 22,000 ohms ± 5 percent, 1/4 watt.

4.2.1.5 Digital voltmeter. The digital voltmeter (DVM) shall nave a voltage accuracy and resolution of 0.1 percent or better from zero to 100 Vdc. Input impedance greater than 10 megohms on all voltage **scales.**

4.2.2 Functional tests.

4.2.2.1 Test connections. The power supplies and resistors shall be connected to the assembly under test in accordance with Figure 1.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:37Z
Check the source to verify that this is the current version before use.

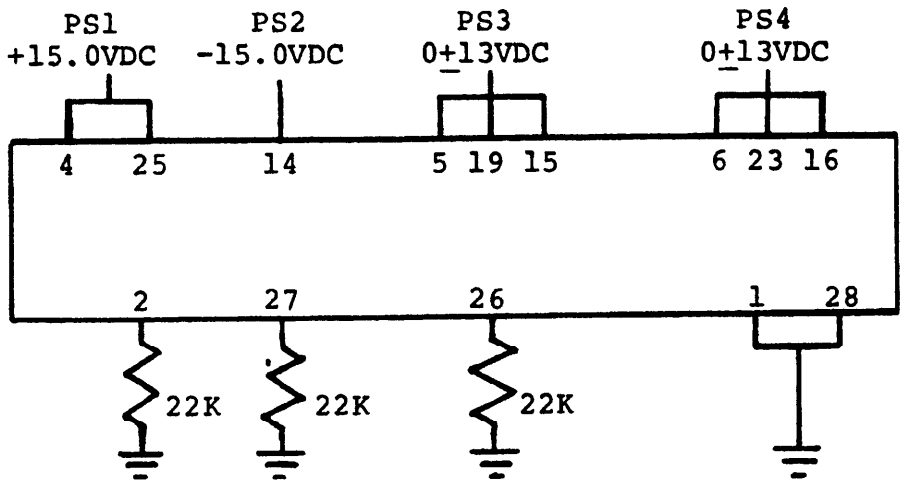


FIGURE 1. Test connections

4.2.2.2 **Test** procedure. Compliance with the requirement of 3.2.1.1.2.1 through 3.2.1.2.3 shall be verified as follows:

a. With the adjustable power supplies set to -13.05 ± 0.05 Vdc, such that PS4 is 0.035 to 0.040 Vdc more negative than PS3, the voltage measured at pins 2, 26, and 27 shall be 11.4 Vdc ± 10 percent.

b. With the adjustable power supplies set to -13.05 ± 0.05 Vdc such that PS4 is 0.035 to 0.040 Vdc more positive than **Ps_3, the voltage measured at pins** 2, 26, **and 27** shall be zero **± 0.5 Vdc.**

c. With the adjustable power supplies set to 0.050 ~ 0.050 Vdc such that PS_4 is 0.035 to 0.040 Vdc more negative than Ps_3, the voltage measured at pins 2, 26, **and** 27 shall be 11.4 Vdc ~+ 10 percent.

d. With the adjustable power supplies set to 0.050 ± 0.050 Vdc such **that PS4 is 0.035 to 0.040 Vdc more positive than PS3, the voltage measured at pins 2, 26, and 27 shall be zero ± 0.5 Vdc.**

e. With the adjustable power supplies set to 13.05 ± 0.05 Vdc such that PS4 is 0.035 to 0.040 Vdc **more** negative than PS3, the voltage **measured at pins 2, 26, and 27 shall be 11.4 ± 10 percent.**

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:37Z
Check the source to verify that this is the current version before use.

* f. With the adjustable power supplies set to 13.05 + 0.05 such that PS4 is 0.035 to 0.040 Vdc more positive than PS3, the voltage measured at pins 2, 26, and 27 shall be zero ± 0.5 Vdc .

5. PREPARATION FOR DELIVERY.

* 5.1 Packaging. Packaging and marking shall be in accordance with the Level A requirements of Packaging Data Sheet (PDS), AMC Form 1029, 9337291.

6. NOTES .

6.1 The notes specified in the general specification MIL-C-50785 are applicable to this specification.

* 6.2 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.3 Orderingtdata. Procurement documents should specify the following:

Title, number, and date of thiS specification.
& Selection of applicable levels of preservation, packaging, and packing.
c. Applicable stock number.
d. Applicable packaging data sheet (see 5.1) .
e. First article sample requirements.

6.4 Changes to previous issue.

*** The outside margins have** been marked **to indicate where additions** to the previous amendment have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies is this notation. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notation and relationship to the last previous issue.

Custodian:
Army - AR

Preparing Activity:
Army - AR

Project No. 1285-A521

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:37Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY CARD

FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army ARRADCOM, ATTN: DRDAR-TST-S
Dover, New Jersey 07801

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:37Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-50785/1B CIRCUIT CARD ASSEMBLY 9337291

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:37Z
Check the source to verify that this is the current version before use.